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Influence of water electrolysis on hydrolysis and methanogenesis stages of anaerobic digestion at room temperature: Kinetic and metabolic analysis

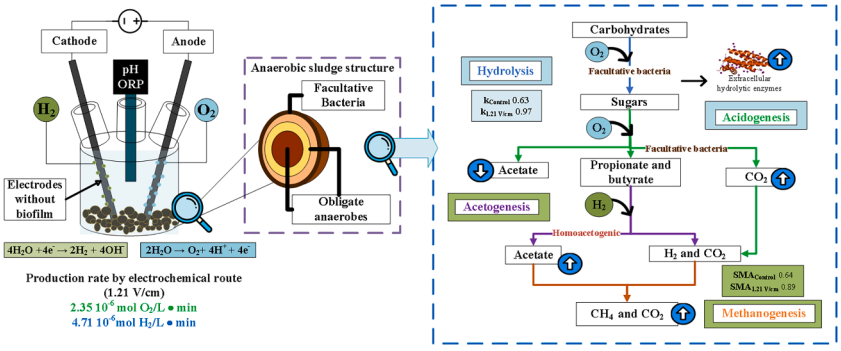
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HIGHLIGHTS

- A low-magnitude electric field improves the AD process by suspended granules.
- RuO_2/Ti favored electrolysis water reactions without biofilm formation.
- The O_2 electrochemically generated accelerates the hydrolysis step in AD up to 72%.
- O_2 stimulates extracellular ORP, modifying anaerobic metabolism.
- H_2 electrochemically generated enhanced the methane yield by up to 39%.

GRAPHICAL ABSTRACT



ARTICLE INFO

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Anaerobic digestion
Oxygen
Facultative bacteria

ABSTRACT

Batch cultures were performed to study hydrolysis and methanogenesis in the presence of an electric field at room temperature (i.e., 23 °C). Kinetic and metabolic analyses using RuO_2/Ti electrodes were carried out in short reaction times to avoid biofilm formation, allowing the evaluation of the effect of O_2 and H_2 produced on anaerobic digestion during the imposition of three electric fields: 1.21, 1.45, and 1.64 V/cm. Results highlighted that at 1.21 V/cm, the electrolysis produced 0.0753 mg O_2/L·min, where facultative microorganisms consumed 21 % oxygen, enhancing the hydrolysis phase by 52 %. Additionally, methane production was noticeably improved with an activity of 0.89 ± 0.02 g COD-CH_4/g VSS·d, meaning 39 % higher than the control. The imposition of an electric field showed promising results since the methanogenic activity at room temperature was very close to the activities observed in conventional reactors at 35 °C.

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1. Introduction

Anaerobic digestion (AD) represents a cost-effective and widely applied biotechnology to treat high-COD wastewater. It is a complex biotechnological process requiring high temperatures (e.g., 35–44 °C) to achieve high methane yields (Mao et al., 2015), so controlling high temperatures in cold climate zones in wastewater treatment plants is economically unfeasible. Therefore, developing technologies to operate AD at low temperatures while maintaining high yields is a real challenge, and this is the main goal of the present work.

Currently, there are some strategies to enhance CH_4 yield, but one of the most recent is the coupling of electrochemical systems, improving COD removal and CH_4 production (Mendoza-Tinoco et al., 2023). For instance, 0.05 to 3.5 V applied on AD promotes electroactive biofilms on the electrode surface, allowing attached bacteria to transfer charges, turning them into redox mediators, thus increasing CH_4 yield (De Vrieze et al., 2018). Koch et al. (2015) imposed a cell potential of 0.2 V, using graphite electrodes at 37 °C, improving CH_4 yield by 27 %. Sun et al. (2020) evaluated an anaerobic reactor at HRT of 24 h, with titanium mesh electrodes (0.05–0.2 V) at 35 °C, observing an increment of methane production by 15 % with shifts in the microbial diversity. The effect of electrochemical reactions in these systems has been little studied, specifically in the granules suspended in the liquid medium. Consequently, the intrinsic production of H_2 and O_2 by the electrolysis of water should be one of the variables to consider since AD syntropy allows microorganisms to couple the O_2 and H_2 produced through electrochemical reactions in their biochemical pathways and, therefore, could regulate AD metabolism. For example, Tartakovsky et al. (2011) observed an AD improvement by applying a cell potential of 2.8–3.5 V for water electrolysis at 35 °C; the authors attributed the higher CH_4 yield to the electrochemical production of O_2 that favored the hydrolytic activity.

Based on the observations mentioned above, it is necessary to develop studies that support the production rate of O_2 and H_2 and their direct effect on each AD stage, mainly in the hydrolysis and methanogenesis, considered the limiting steps. Therefore, this work aims to evaluate the influence of oxygen production by water electrolysis in hydrolysis and methanogenesis activities using only granular sludge previously stabilized at room temperature in batch cultures.

2. Materials and methods

2.1. Inoculum and culture medium

Anaerobic sludge from a candy-manufacturing wastewater treatment plant was stabilized in a laboratory-scale Upflow Anaerobic Sludge Blanket (UASB) reactor with an operating volume of 2 L. It was fed with the following chemical medium: (in g/L), sucrose (1.5), NH_4CI (0.85), K_2HPO_4 (0.234), KH_2PO_4 (0.136), $MgCl{\cdot}6H_2O$ (0.084), $NaHCO_3$ (2), and yeast extract (0.34). A trace element solution (0.5 mL/L) was added, with the following composition (in g/L), EDTA (0.5), $ZnSO_4{\cdot}7H_2O$ (0.4), $CoCl_2{\cdot}6H_2O$ (0.25), $MnCl_2$ (0.6), $CuSO_4{\cdot}5H_2O$ (0.25), $Na_2MoO_4{\cdot}2H_2O$ (0.2), $NiCl_2{\cdot}6H_2O$ (0.2), $MgCl_2$ (0.5), $FeSO_4$ (0.1), and $CaCl_2{\cdot}2H_2O$ (0.1).

2.2. Characterization of the bioelectrochemical system for batch cultures

Electrochemical characterization was performed in glass electrochemical reactors (two-electrode cells) with 130 mL of liquid and 10 mL of head space. The electrodes were constructed in Ti plates coated with RuO_2 (RuO_2/Ti), whose dimensions were 7.5 cm long, 0.6 cm wide, and 0.1 cm thick; the submerged area of the electrodes was 9.16 cm^2, and the separation between electrodes was 2 cm. The electric field is associated with the cell potential where two Ti/RuO_2 electrodes are immersed in the granules and culture medium. A low electric current is imposed, and the cell potential developed is monitored with the imposition time. Electric field vs. current density curve has been used to find an operating interval in which a resistive behavior was observed to ensure no chemical alterations in the culture chemical composition (Gómez-Flores et al., 2020). A high-precision power supply (BK PRECISIONS Model 9120A, Taiwan) was used to apply the electric current. A voltammetric study was carried out with a mineral medium containing sucrose and acetate. A voltammogram from −1 to 1.7 V with a scan potential of 10 mV/s was performed (in a three-cell electrode) by using mercury/mercurous sulfate/saturated sulfate (SSE) and a carbon cylinder as reference and counter electrodes, respectively.

2.3. Characterization of the anaerobic digestion stages applying an electric field

Batch cultures were inoculated with 2 g VSS/L previously acclimatized in the UASB. Experimental units were bubbled with N_2 for 3 min to ensure an oxygen-free environment. Initially, batch cultures (without an electric field) were carried out as a control assay. Subsequently, the effect of the electric field on hydrolysis and methanogenesis was evaluated by applying three magnitudes (1.21, 1.45, and 1.64 V/cm). In the hydrolysis culture, sucrose was used as a carbon source at 0.9 g/L, where sucrose consumption was monitored over time. In the methanogenic culture, 1.5 g/L acetic acid was added; where methane production was plotted, and the specific methanogenic activity (SMA) was determined. Origin Pro Software was used to calculate the kinetic parameters using the Gompertz Model. A single-factor analysis of variance (ANOVA) was performed to compare the experimental results obtained in the presence and absence of an electric field. The significance value (α) was 5 % using the data analysis add-in of Microsoft Excel software.

2.4. Determination of O_2 production and facultative activity

In batch cultures, the dissolved oxygen (DO) concentration was quantified in the presence and absence of an electric field; this consisted of applying the three selected electric field magnitudes. Abiotic (without sludge) and biotic (with 2 g VSS/L of sludge) experiments were performed with constant stirring at 100 rpm for five hours. The O_2 consumption capacity of the sludge was determined in independent experiments in which air was supplied by bubbling until a pseudo-stationary state was reached; once a constant concentration was reached, aeration was stopped, and the DO consumption profile was monitored. The consumption kinetics were established and adjusted to a Monod-type model to determine the O_2 consumption capacity of the facultative microorganisms.

2.5. Analytical methods

pH was measured with a multiparameter (Orion Star™ A215, Indonesia). ORP was measured with a selective electrode (Cole-Parmer EW-27009-30) connected to a pH/ORP controller meter (Eutech Instruments 800, Singapore). Phenol–sulfuric acid and DNS techniques were used to quantify total and reducing sugars, respectively. Volatile fatty acids were quantified in a gas chromatograph (Agilent Technologies 7820A, Shanghai) with a flame ignition detector (FID), using N_2 as a carrier gas, with a heating ramp from 80 °C to 200 °C, with increments of 25 °C/min, with an Agilent J&W DB-FFAP column. The biogas composition was determined using helium as the carrier gas with a gas chromatograph (Gow-Mac 580 GC, U.S. A) with a thermal conductivity detector (TCD). The operating conditions were as follows: T_{column} = 40 °C, $T_{injector}$ = 90 °C, $T_{detector}$ = 90 °C, detector current: 120 mA. The DO was quantified with the sensor based on the phenomenon of photoluminescence (WTW™ FDO 925, Germany).

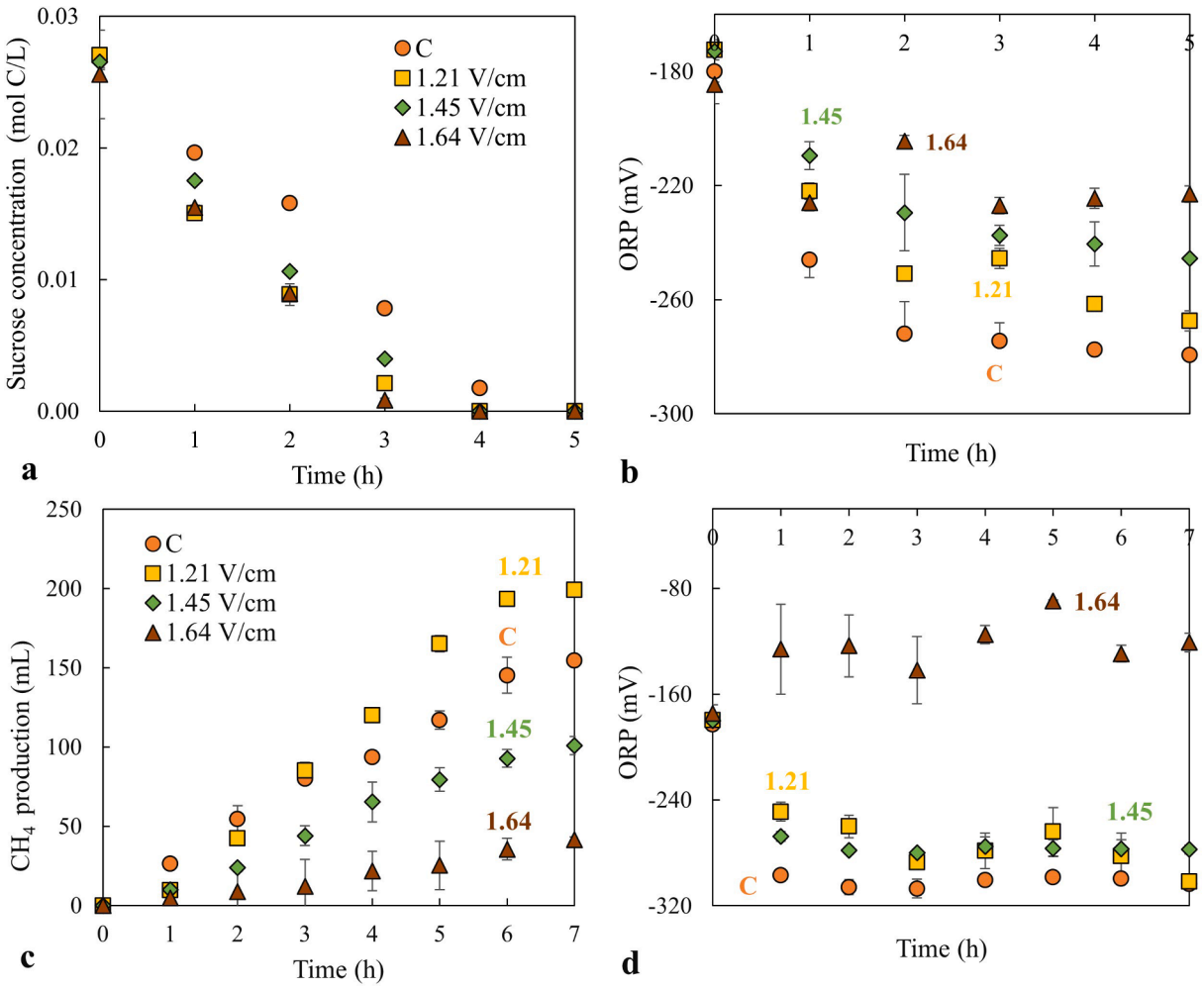


Fig. 1. Comparison of hydrolysis and methanogenic profiles and ORP behavior at 23 °C in the absence of the electric field (C) and the presence of different electric fields (marked in the figure: 1.21, 1.42, and 1.64 V/cm). a) Sucrose consumption, b) ORP during the hydrolysis, c) CH_4 production, and d) ORP during methanogenesis. Mol C corresponds to total concentration of carbon in the sacarose.

3. Results and discussion

3.1. Characterization of the bioelectrochemical system

Cell potential versus applied current density curves were constructed to determine a current density interval in which the media behaves like an electrical resistance (see Supplementary material). Electrical resistance behavior allows the operation to ensure that the electrochemical reactions on the surface of the electrodes will not alter the chemical composition of the medium (Gómez-Flores et al., 2020). The resistive behavior was found by applying an interval of 0.54 to 1.63 mA/cm^2 (Supplementary material); the electric field magnitudes applied to the cell were 1.21, 1.45 and 1.64 V/cm.). The voltammogram obtained in a mineral medium containing sucrose and acetate showed neither reduction nor oxidation of any chemical component in the culture medium; only water electrolysis reactions occurred at the anodic and cathodic potential barriers (see Supplementary material).

3.2. Effect of the electric field on the hydrolysis stage

The consumption profile of sucrose in the control study was consumed at five hours, while the studies with the imposition of 1.21, 1.45, and 1.64 V/cm reached it at approximately 4 h (Fig. 1a). The electric field effect was reflected in the hydrolysis efficiencies, which when compared at 3 h, the control had a hydrolysis efficiency of 68 ± 1.42 %. In contrast, in the applied electric fields of 1.21, 1.45, and 1.64 V/cm, the hydrolysis efficiencies obtained were 91 ± 0.28 %, 83 ± 0.20 %, and 93 ± 1.18 %, respectively.

Table 1
Influence of the magnitude of the electric field on the kinetic parameters of hydrolysis and methanogenesis stages.

Stages	Electric field (V/cm)	k (h^{-1})	Characteristic time (h)
Hydrolysis	0	0.63 ± 0.00^a	1.59^a
	1.21	0.97 ± 0.02^b	1.03^b
	1.45	0.90 ± 0.00^c	1.12^c
	1.64	1.08 ± 0.01^d	0.93^d
Methanogenesis	0	0.40 ± 0.01^a	2.53^a
	1.21	0.55 ± 0.02^a	1.83^a
	1.45	0.50 ± 0.10^a	2.08^a
	1.64	0.46 ± 0.04^a	2.21^a

*The difference between letters indicates that they are significantly different.

The kinetic analysis revealed that the reaction constant increased compared to the control by 52, 42, and 72 % for electric field magnitudes of 1.21, 1.45, and 1.64 V/cm, respectively (Table 1). These results indicated that the characteristic time of the hydrolysis stage decreases with increases in the electric field, which could improve the AD's overall performance. Tartakovsky et al. (2011) suggested that microaerobic

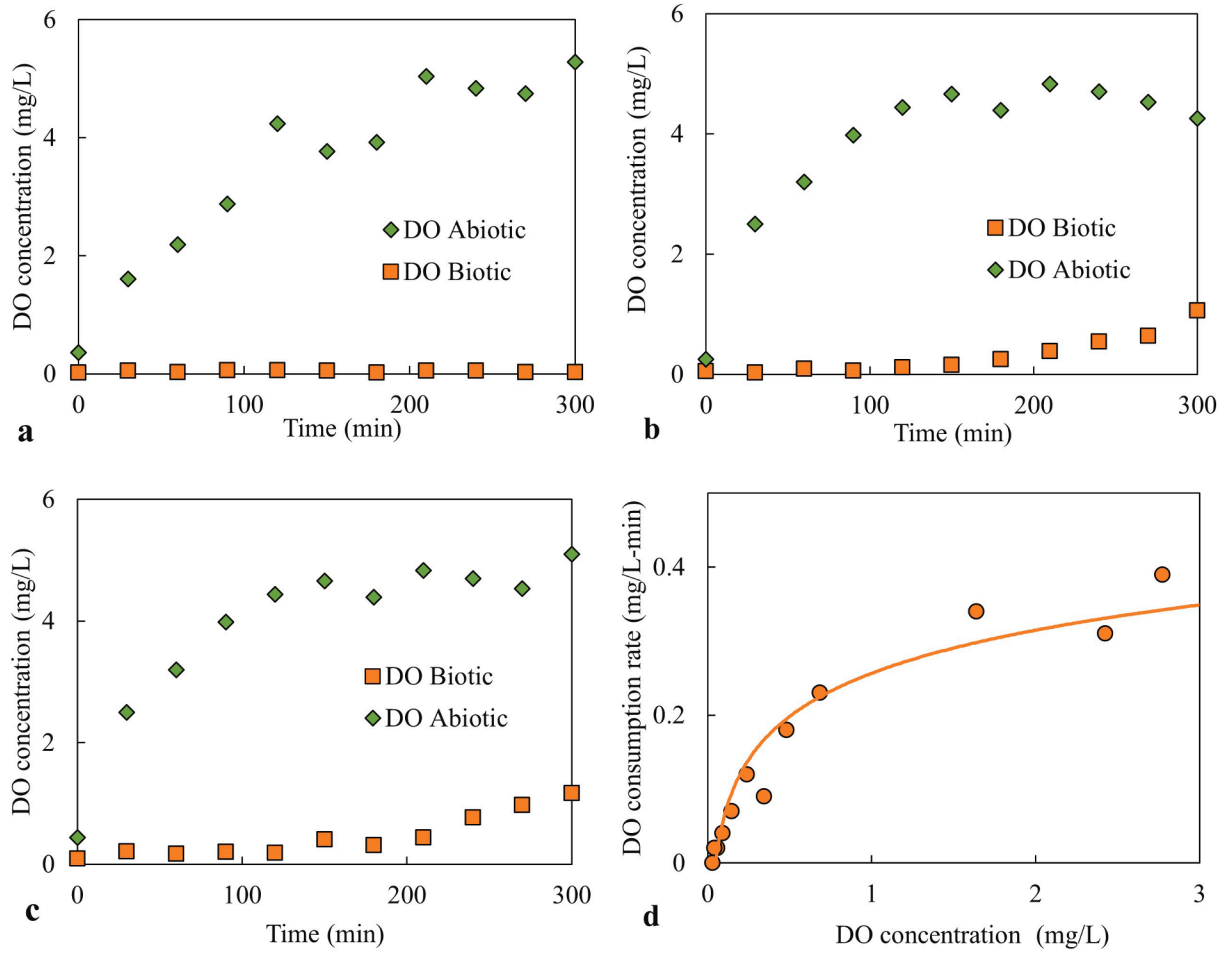


Fig. 2. Comparison of dissolved oxygen (DO) concentration in the hydrolysis stage at 23 °C, applying different electric fields in a biotic and abiotic medium. a) 1.21 V/cm, b) 1.45 V/cm, c) 1.64 V/cm, d) kinetic analysis of oxygen consumption, based on the Monod-type model (Continuous line), obtained in the biotic system (previously saturated with O_2) in the absence of an electric field.

conditions by water electrolysis might favor the proliferation of facultative bacteria with better hydrolytic activities due to aerobic routes. On the other hand, Zhang et al. (2021) indicated that microaerobic conditions might increase extracellular enzyme production, speeding up the organic matter hydrolysis.

In abiotic experiments, three electric field magnitudes were applied to support the previous proposals, reaching similar oxygen saturation concentrations (Fig. 2a). However, the production rate increased proportionally to the magnitude of the electric field: 0.0753, 0.0822, and 0.0943 mg O_2/L·min for 1.21, 1.45, and 1.64 V/cm, respectively (Fig. 2a-c). In biotic experiments, DO is consumed almost entirely, in which the lowest concentration of DO was obtained at 1.21 V/cm (Fig. 2a). The data obtained in the biotic system (previously saturated with O_2) in the absence of the electric field (Fig. 2d) showed a maximum oxygen consumption rate of 0.484 ± 0.049 mg O_2/L·min with an affinity constant of 0.877 ± 0.222 mg/L. Based on these values, the oxygen consumption rates were estimated at 0.016 ± 0.002, 0.123 ± 0.020, and 0.154 ± 0.012 mg O_2/L·min for the applied electric fields of 1.21, 1.45, and 1.64 V/cm, respectively. These results show how, from 200 min, the DO begins to accumulate in the aqueous stage with 1.45 and 1.64 V/cm since the capability of microorganisms to consume O_2 is overwhelmed (Fig. 2b-c). At the same time, when 1.21 V/cm was applied, facultative bacteria maintained a microaerobic environment with a concentration of 0.04 ± 0.01 mg/L (Fig. 2a). Therefore, it can be concluded that the electric field decreases the characteristic time for the hydrolysis stage, which is attributed to the fact that the electrochemically generated oxygen favors the metabolism of facultative bacteria.

On the other hand, in all batch cultures, pH values were within the range of 7.2 to 7.7, an acceptable value for anaerobic digestion. The ORP in the hydrolysis stage displayed less negative values in the presence of the electric field, and this behavior was proportional to the increase in electric field magnitude (Fig. 1b). The profiles were stabilized from hour two and remained constant until hour five; the control presented a value of −272 ± 11.3 mV, while the electric field presented values of −251 ± 0.7, −229 ± 3.5 and −226 ± 2.8 mV for electric field magnitudes of 1.21, 1.45 and 1.64 V/cm, respectively.

3.3. *Effect of the electric field on the methanogenesis stage*

Batch cultures fed with acetic acid to evaluate methanogenic activity showed that the three tests performed with an electric field display an adaptation stage at one hour with a methane production of 12.5 ± 2.5 mL (Fig. 1c). In contrast, the control showed a higher methane production in the first hour of 32.5 mL ± 2.1. Methanogens are the most susceptible to oxygen because they cannot synthesize the superoxide dismutase enzyme that aerobic organisms use to neutralize reactive oxygen species. However, in the presence of O_2 to overcome oxidative stress, some methanogenic species may have intrinsic tolerance to oxygen exposure as they adapt to environmental conditions and can even produce superoxide dismutase, which could explain how these experiments showed this adaptive stage (Magdalena et al., 2022).

On the other hand, the control presented a SMA value of 0.64 ± 0.003 g COD-CH_4/g VSS·d; this agrees with data reported in the literature (Gonzalez-Tineo et al., 2020). At the same time, assays in the

presence of an electric field were 0.89 ± 0.02, 0.44 ± 0.05, and 0.13 ± 0.001 g COD-CH_4/g VSS·d for magnitudes of electric fields of 1.21, 1.45, and 1.64 V/cm, respectively; the batch cultures performed at 1.21 V/cm presented an increase of 39 % compared to the control.

Most known methanogenic archaea are mesophilic and have an optimum temperature of 35 °C, diminishing the SMA when temperature is dropped. For example, Kayranli and Ugurlu (2012) using acetate obtained a SMA of 0.74 g COD/g VSS·d at 25 °C, 0.70 g COD/g VSS·d at 15 °C, and 0.68 g COD/g VSS-d at 10 °C, in comparison with the research of Soto et al. (1993), who observed an SMA between 1.078 and 0.975 g COD/g VSS·d at 35 °C. In the present study, the SMA exceeds the value reported at 25 °C, meaning that the electric field enhances this biological process, competing successfully with conventional AD at 35 °C.

Finally, 1.21 and 1.45 V/cm of electric fields did not affect the reducing conditions since the ORP was similar to the culture control, maintaining values of −277.33 ± 2.1 to −302.5 ± 2.8 mV from hours two to five (Fig. 1d). However, at 1.64 V/cm, the ORP was less negative regarding the control, with an average value of −120.08 ± 7.2 from hours two to seven. Nghiem et al. (2014) reported that microaeration in an anaerobic digester changed the ORP to less negative values, from −485 mV to −270 mV.

4. Conclusions

By conducting water electrolysis caused by a low-magnitude electric field at 23 °C, the outcome of anaerobic digestion was strongly impacted. The micro-aerobic conditions enhanced the hydrolysis rate of sucrose at any electric field, thus improving the organic matter removal. The microbial methanogenic activity was positively stimulated at the lowest electric field since acetate consumption was increased with a concomitantly enhanced methane production; real matrix will be used to validate the positive effect of the electric field. These findings open an alternative to significant development using a controlled low-magnitude electric field to redirect the desired products without temperature control.

CRediT authorship contribution statement

Tania Paola Mendoza-Tinoco: Data curation, Formal analysis, Investigation, Methodology, Writing – original draft. **Ulises Durán-Hinojosa:** Conceptualization, Data curation, Formal analysis, Methodology, Resources, Supervision, Writing – original draft, Writing – review & editing. **Víctor Sánchez-Vázquez:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing. **María del Carmen Fajardo-Ortiz:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Ricardo Beristain-Cardoso:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Ignacio González:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2023.130096.

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